Reports Module – Technical Documentation

> **Project:** TaskFlow – Intelligent Task Allocation & Tracking System **Module:** Reports · **Child Modules:** Reports **Version:** 1.0 · **Last Updated:** 31-Oct-2026 **Backend:** ASP.NET Core Web API (.NET 10) · **Frontend:** Angular · **Database:** SQLite (EF Core) **Author:** Jojo George (jojogeorge3344@gmail.com) © 2026 Jojo George. All rights reserved.

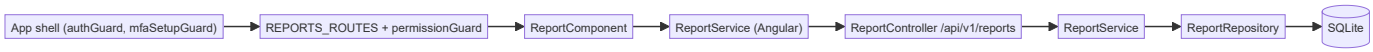
> ✅ Target framework confirmed: `net10.0` (.NET 10), from `backend/src/TaskFlow.Api/TaskFlow.Api.csproj`.

1. Technical Overview

The Reports module is a read-only reporting slice for BRD Screen 17. It has no tables of its own and reads the `Tasks`, `Users`, `UserProjectAllocations` and `Projects` tables. It has one child module, **06.1 Reports**, which exposes two GET endpoints under `api/v1/reports` (summary totals and CSV export).

Data is **organization-wide** for every caller who holds `reports.view`. No per-user scope is applied (design note D3 in `ReportService`). The Angular `ReportComponent` is a standalone page loaded under the authenticated shell.

2. Architecture Flow



3. Backend File Structure

Layer	Path	Child
Controller	`TaskFlow.Api/Controllers/Reports/`	06.1
Service	`TaskFlow.Api/Services/Reports/`	06.1
Repository	`TaskFlow.Api/Repositories/Reports/`	06.1
DTOs / Validators	`TaskFlow.Api/Dtos/Reports/`, `TaskFlow.Api/Dtos/Reports/Validators/`	06.1
Entities / Enums (read only, shared)	`TaskFlow.Api/Models/Entities/`, `TaskFlow.Api/Models/Enums/`	Shared
Authorization (shared)	`TaskFlow.Api/Authorization/Policies/`, `TaskFlow.Api/Common/Security/` (`RequirePermissionAttribute.cs`, `PermissionPolicy.cs` prefix `perm:`, `PermissionKeys.cs`, `PermissionCatalog.cs`)	Shared
Cross-cutting (shared)	`TaskFlow.Api/Filters/ValidationFilter.cs`, `TaskFlow.Api/Middleware/GlobalExceptionHandler.cs`, `TaskFlow.Api/Authorization/Auditing/DeniedActionAuditHandler.cs`	Shared
DI registration	`TaskFlow.Api/Config/ServiceRegistration.cs`, `RepositoryRegistration.cs`	06.1

File-level detail is in `06-Reports/06.1-Reports.md`.

4. Frontend File Structure

Type	Path	Child
Feature folder	`src/app/modules/reports/` (`pages/`, `services/`, `models/`, `reports.routes.ts`)	06.1
Shell route	`src/app/routes/app.routes.ts` (`authGuard`, `mfaSetupGuard`)	Shared
Menu	`src/app/layouts/nav-items.ts` ("Reports", icon `chart`, route `/reports`)	06.1
Permission constants	`src/app/shared/auth/permissions.ts` (`ReportsView`, `ReportsExport`)	06.1
Shared utilities	`src/app/shared/utils/`, `src/app/shared/services/toast.service.ts`, `shared/components/icon/`	Shared

5. Child Modules

#	Child Module	Main endpoints (count)	Key permissions	Document
6.1	Reports	`GET /api/v1/reports`, `GET /api/v1/reports/export` (2)	`reports.view`, `reports.export` (plus `projects.view` for the Project dropdown)	`06.1-Reports.md`

6. Database Design

The module owns no tables. All reads are made by child 06.1.

Table	Owner	Read by	Use
`Tasks`	Monitoring (Task Management)	06.1 Reports	Filters, totals, export rows
`Users`	Management (Employee Management)	06.1 Reports	Assignee name (soft-delete filter bypassed)
`UserProjectAllocations`	Management (Project Management)	06.1 Reports	Project filter, CSV `Project` column
`Projects`	Management (Project Management)	06.1 Reports	Project names

Relationships

- `Tasks.AssignedToUserId → Users.Id` (FK, Restrict)
- `Tasks.ProjectId → Projects.Id` (FK, Restrict). Not read by this module.
- `UserProjectAllocations.UserId → Users.Id` (FK, Cascade)
- `UserProjectAllocations.ProjectId → Projects.Id` (FK, Cascade)

Enum storage: `TaskStatus` = ToDo 0, InProgress 1, Blocked 2, Completed 3, Cancelled 4, InQA 5. `TaskPriority` = Low 0, Medium 1, High 2, Critical 3. They are stored as integers and sent over JSON as names (`JsonStringEnumConverter` in `CrossCuttingRegistration.cs`).

Column-level detail is in the child document.

7. Security & Authorization

- **Authentication:** JWT bearer access token (`Jwt:AccessTokenMinutes`) plus a refresh-token cookie (`Jwt:RefreshTokenDays`).
- **Authorization:** Permission policies named `perm:` + key (`PermissionPolicy.NameFor`). Permission claims are refreshed from the database on every request. Refusals are logged by `DeniedActionAuditHandler` (user id, method, path, policy) and return the default 403.
- **Catalog (`PermissionCatalog.cs`):** module `Report` , screen `Reports` . `reports.view` ("View reports", action `View` , `GrantedToManager: true`); `reports.export` ("Export reports", action `Export` , `GrantedToManager: true` , `GrantFrom: reports.view` , `Requires: reports.view`). Administrator is granted every key on every start (`AccessControlSeeder`). Employee has neither key by default.
- **Data scope: Organization** for every holder of `reports.view` (D3, "matching the dashboards' scoping").

Role	`reports.view`	`reports.export`
Administrator	Granted	Granted
Manager	Granted	Granted
Employee	Denied	Denied
Custom Role	Configurable	Configurable

Key → endpoint → UI mapping is in the child document.

8. Validations & Error Handling

Shared handling used by the child:

Rule	Layer	HTTP Code	Error Message
Not signed in / expired token	Authentication	401	–
Missing permission	Policy	403	Forbidden (default handler, audited)
Unhandled fault	`GlobalExceptionHandler`	500	Something went wrong. Please try again.

Errors are returned as `application/problem+json` by `GlobalExceptionHandler` . Validation failures are returned as `ValidationProblemDetails` by `ValidationFilter` . Child-specific rules are in `06.1-Reports.md` .

9. Integrations / Side Effects

- **Notifications / Email / Activity Log:** None in this module. Only refused calls are logged, by `DeniedActionAuditHandler` .
- **AI Assistant:** No report tool is exposed. The help article `Services/Assistant/Knowledge/reports-and-workload.md` describes the screen to users.

10. Configuration

Key (`appsettings.json`)	Purpose	Default
`ConnectionStrings:Default`	SQLite database used by `AppDbContext`	`<secret>`
`Jwt:AccessTokenMinutes`	Access-token lifetime for calling the report APIs	30
`Jwt:RefreshTokenDays`	Refresh-token cookie lifetime	7

The module has no configuration section of its own. Child-specific settings and code constants are in the child document.

11. Testing

Test Type	File / Location	Covers
Integration	`tests/TaskFlow.IntegrationTests/Common/PermissionAuthorizationTests.cs`	Every catalog key, `reports.view` included, has a policy that needs its claim. A custom role holding only `reports.view` gets 200 on `GET /api/v1/reports` and 403 on workload, admin dashboard and users.
Integration	`tests/TaskFlow.IntegrationTests/Common/ModuleBasedPermissionEndpointsTests.cs`	`GET /api/v1/reports/export` returns 403 without `reports.export` and is reached with `reports.export` plus `reports.view`.
Unit	`tests/TaskFlow.UnitTests/Settings/RoleConfiguration/PermissionCatalogTests.cs`	`reports.view` and `reports.export` sit in the "Report" module. A role that held old report access keeps `reports.export` after the upgrade. The Employee role does not hold `reports.view`.
Front end unit	`src/app/routes/app.routes.spec.ts`	The `reports` route uses `permissionGuard` with `reports.view` and loads `ReportComponent`.
Front end unit	`src/app/layouts/nav-items.spec.ts`	Reports is a top-level sidebar item for Administrator and Manager and is not shown to Employee.

> Report-specific tests are listed in the child document.

12. Known Limitations / Future Enhancements

Child-specific gaps are listed in each child document.

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